Milestone1_simulate

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Using simulated data modelled on OECD unemployment statistics, this project plans to compare tertiary-educated foreign-born and native-bornwomen across more and less developed regions, and by age at arrival.

Unemployment among Highly Educated Immigrant Women by Destination and Age at Arrival

Immigrants with a university degree are often expected to secure stable positions in the labor market, yet a degree does not guarantee the same outcome across nations. Immigrant women may face disadvantages that combine gender and migration status, and the strength of these disadvantages may depend on where they settle and on when they arrived. Two factors seem especially worth examining. First, the destination country: as_____ ,I expect highly educated foreign-born women to experience lower unemployment in Northern America, Oceania, Western Europe and Northern Europe, which I group as "more developed" regions, than in Latin America, Southern Europe and Eastern Europe, which I group as "less developed" regions. Second, age at arrival: women who arrived before age 16 were likely to have incorporated and completed much of their schooling in the destination country, so I expect their unemployment to be lower than that of women who arrived at age 16 or older, whose qualifications may have been earned elsewhere and who may have had less time to become familiar with the destination country.

A low unemployment rate among immigrants may simply reflect a strong labor market that benefits everyone. To separate the two, I also compare each foreign-born group with native-born women of the same region and education level, and examine the gap between them (foreign-born minus native-born unemployment rate). If immigrant women are better off in more developed regions for reasons beyond general labor market conditions, I expect this gap to be smaller in more developed regions than in less developed ones.

Using OECD data on unemployment rates by educational attainment, migration status, age group and gender, I describe how these two factors relate to unemployment among tertiary-educated foreign-born women.

```r
library(tidyverse)
library(tinytex)

set.seed(916)

#Simulate the structure of the real data, based on the data of OECD
country_region <- tibble(
  country = c("Canada", "United States",
              "Mexico", "Costa Rica", "Colombia", "Chile",
              "Australia", "New Zealand",
              "Austria", "Belgium", "France", "Germany", "Luxembourg",
              "Netherlands", "Switzerland",
              "Estonia", "Ireland", "Latvia", "Lithuania", "Norway",
              "Sweden", "United Kingdom",
              "Greece", "Italy", "Portugal", "Slovenia", "Spain",
              "Bulgaria", "Czechia", "Hungary", "Poland", "Romania",
              "Slovak Republic"),
  region  = c(rep("Northern America", 2),
              rep("Latin America", 4),
              rep("Oceania", 2),
              rep("Western Europe", 7),
              rep("Northern Europe", 7),
              rep("Southern Europe", 5),
              rep("Eastern Europe", 6))
)

head(country_region)
```

```
# A tibble: 6 x 2
  country       region
  <chr>         <chr>
1 Canada        Northern America
2 United States Northern America
3 Mexico        Latin America
4 Costa Rica    Latin America
5 Colombia      Latin America
6 Chile         Latin America
```

```r
#
n_countries <- length(country_region$country)   # 33
```

```r
simulated_ppl <- tibble(
  country = rep(country_region$country, each = 18),
  sex = rep(rep(c("Female", "Male"), each = 9), times = n_countries),
  education = rep(rep(c("Below upper secondary",
                        "Upper secondary / post-secondary non-tertiary",
                        "Tertiary"), each = 3),
                  times = 2 * n_countries),
  birth_place = rep(c("Foreign-born", "Foreign-born", "Native-born"),
                    times = 6 * n_countries),
  arrival_age = rep(c("Arrived before age 16", "Arrived at age 16 or older",
                      "Not applicable"),
                    times = 6 * n_countries)
)

head(simulated_ppl)
```

```
# A tibble: 6 x 5
  country sex    education                                birth_place arrival_age
  <chr>   <chr>  <chr>                                    <chr>       <chr>
1 Canada  Female Below upper secondary                    Foreign-bo~ Arrived be~
2 Canada  Female Below upper secondary                    Foreign-bo~ Arrived at~
3 Canada  Female Below upper secondary                    Native-born Not applic~
4 Canada  Female Upper secondary / post-secondary non-t~ Foreign-bo~ Arrived be~
5 Canada  Female Upper secondary / post-secondary non-t~ Foreign-bo~ Arrived at~
6 Canada  Female Upper secondary / post-secondary non-t~ Native-born Not applic~
```

```r
#add region and development group
simulated_groups <-
  simulated_ppl |>
  left_join(country_region, by = "country") |>
  mutate(
    development_group = case_match(
      region,
      c("Northern America", "Oceania", "Western Europe", "Northern Europe") ~ "More developed",
      c("Latin America", "Southern Europe", "Eastern Europe") ~ "Less developed",
      .default = NA_character_
    )
  )

head(simulated_groups)
```

```
# A tibble: 6 x 7
```

```
  country sex    education      birth_place arrival_age region development_group
  <chr>   <chr>  <chr>          <chr>       <chr>       <chr>  <chr>
1 Canada  Female Below upper s~ Foreign-bo~ Arrived be~ North~ More developed
2 Canada  Female Below upper s~ Foreign-bo~ Arrived at~ North~ More developed
3 Canada  Female Below upper s~ Native-born Not applic~ North~ More developed
4 Canada  Female Upper seconda~ Foreign-bo~ Arrived be~ North~ More developed
5 Canada  Female Upper seconda~ Foreign-bo~ Arrived at~ North~ More developed
6 Canada  Female Upper seconda~ Native-born Not applic~ North~ More developed
```

```
#unemployment rate
simulated_rates <-
  simulated_groups |>
  mutate(
    unemployment_rate = sample(seq(1, 15, by = 0.1),
                               size = n(),
                               replace = TRUE)
  )

head(simulated_rates)
```

```
# A tibble: 6 x 8
  country sex    education      birth_place arrival_age region development_group
  <chr>   <chr>  <chr>          <chr>       <chr>       <chr>  <chr>
1 Canada  Female Below upper s~ Foreign-bo~ Arrived be~ North~ More developed
2 Canada  Female Below upper s~ Foreign-bo~ Arrived at~ North~ More developed
3 Canada  Female Below upper s~ Native-born Not applic~ North~ More developed
4 Canada  Female Upper seconda~ Foreign-bo~ Arrived be~ North~ More developed
5 Canada  Female Upper seconda~ Foreign-bo~ Arrived at~ North~ More developed
6 Canada  Female Upper seconda~ Native-born Not applic~ North~ More developed
# i 1 more variable: unemployment_rate <dbl>
```

```
simulated_rates |> summarise(min_rate = min(unemployment_rate),
                              max_rate = max(unemployment_rate))
```

```
# A tibble: 1 x 2
  min_rate max_rate
     <dbl>    <dbl>
1        1     14.9
```

```r
simulated_data <-
  simulated_rates |>
  select(region, development_group, sex, education, birth_place, arrival_age, unemployment_ra
```

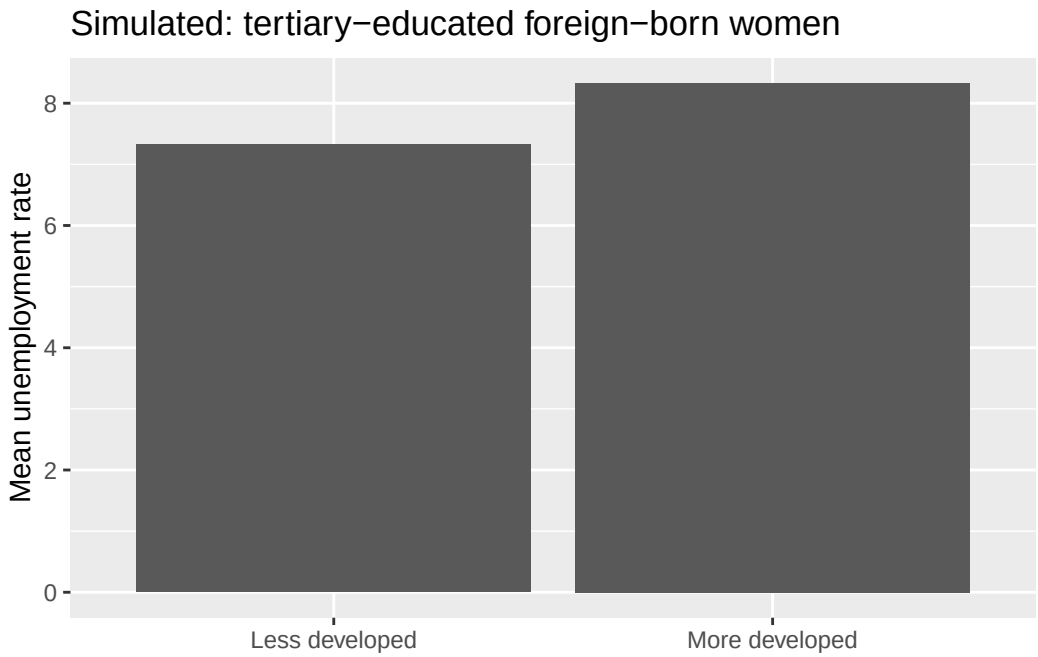
```r
women_tertiary_simulated <-
  simulated_data |>
  filter(sex == "Female",
         education == "Tertiary") |>
  drop_na(development_group, unemployment_rate) |>
  mutate(comparison_group = if_else(birth_place == "Native-born",
                                    "Native-born",
                                    arrival_age))

#  CHANGED: only foreign-born women for the first mean
mean_by_group <-
  women_tertiary_simulated |>
  filter(birth_place == "Foreign-born") |>
  group_by(development_group) |>
  summarise(mean_rate = mean(unemployment_rate))

mean_by_group
```

```
# A tibble: 2 x 2
  development_group mean_rate
  <chr>                <dbl>
1 Less developed        7.33
2 More developed        8.33
```

```r
mean_by_group |>
  ggplot(aes(x = development_group, y = mean_rate)) +
  geom_col() +
  labs(x = NULL,
       y = "Mean unemployment rate",
       title = "Simulated: tertiary-educated foreign-born women")
```




```
mean_by_group |>
  ggplot(aes(x = development_group, y = mean_rate)) +
  geom_col() +
  labs(x = NULL,
       y = "Mean unemployment rate",
       title = "Simulated: tertiary-educated foreign-born women")
```




```
women_tertiary_simulated |>
  filter(birth_place == "Foreign-born") |>                    #   CHANGED (added)
  group_by(region, development_group) |>
  summarise(mean_rate = mean(unemployment_rate), .groups = "drop") |>
  ggplot(aes(x = region, y = mean_rate, fill = development_group)) +
  geom_col() +
  coord_flip() +
  labs(x = NULL,
       y = "Mean unemployment rate",
       fill = "Group",
       title = "Simulated: tertiary-educated foreign-born women, by region")
```



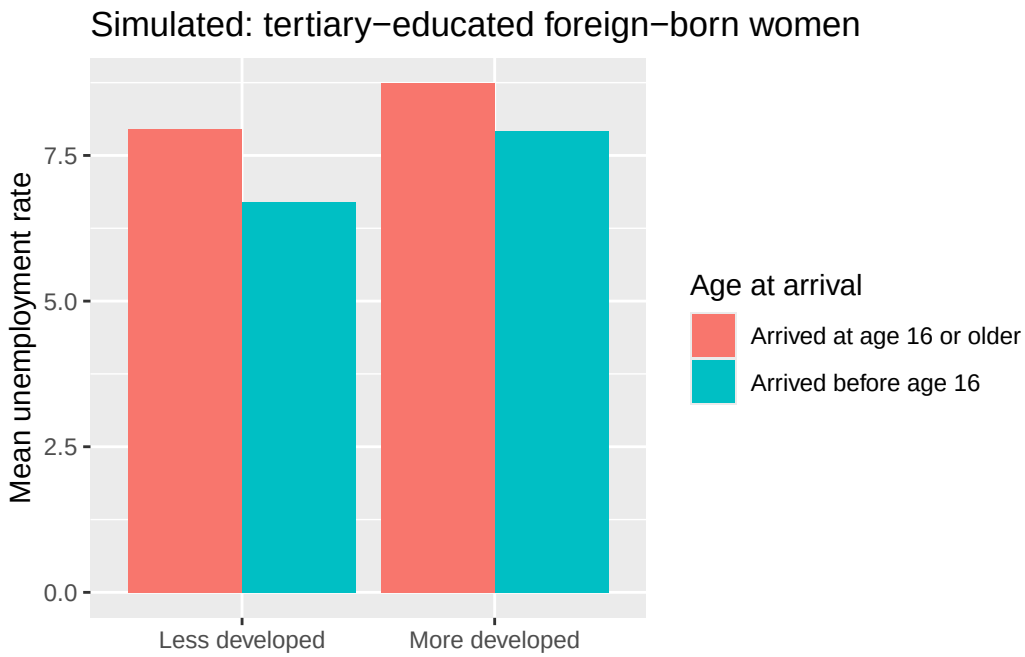

```
mean_by_group_arrival <-
  women_tertiary_simulated |>
  filter(birth_place == "Foreign-born") |>
  group_by(development_group, arrival_age) |>
  summarise(mean_rate = mean(unemployment_rate),
            .groups = "drop")

mean_by_group_arrival
```

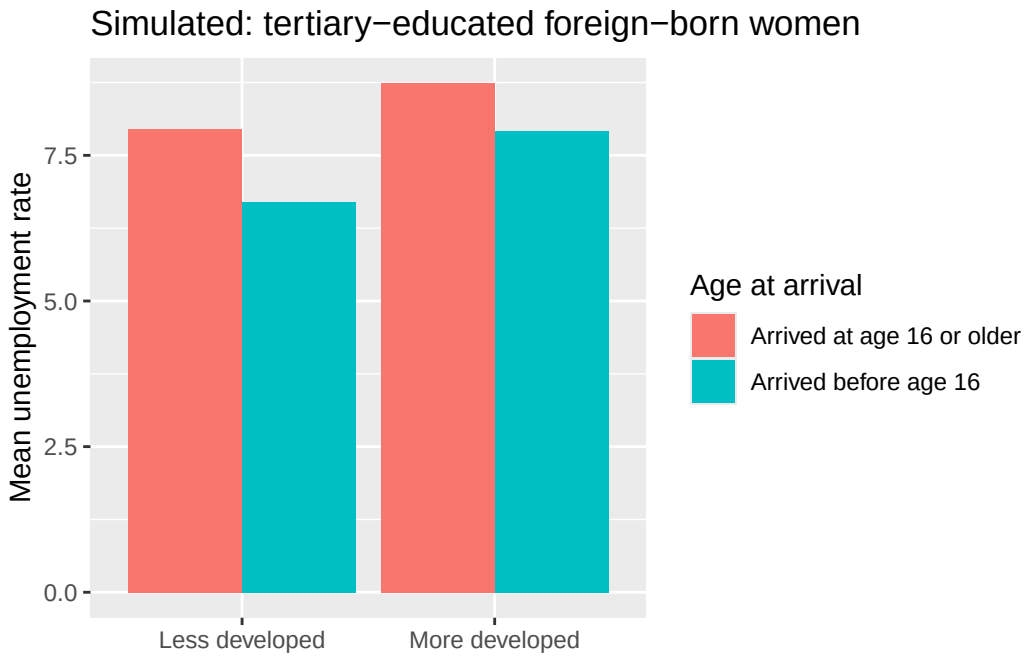
```
# A tibble: 4 x 3
  development_group arrival_age                mean_rate
  <chr>             <chr>                          <dbl>
1 Less developed    Arrived at age 16 or older      7.95
2 Less developed    Arrived before age 16           6.7
3 More developed    Arrived at age 16 or older      8.74
4 More developed    Arrived before age 16           7.92
```

```
mean_by_group_arrival |>
  ggplot(aes(x = development_group, y = mean_rate, fill = arrival_age)) +
  geom_col(position = "dodge") +
```

```
  labs(x = NULL,
       y = "Mean unemployment rate",
       fill = "Age at arrival",
       title = "Simulated: tertiary-educated foreign-born women")
```




```
mean_by_group_arrival |>
  ggplot(aes(x = development_group, y = mean_rate, fill = arrival_age)) +
  geom_col(position = "dodge") +
  labs(x = NULL,
       y = "Mean unemployment rate",
       fill = "Age at arrival",
       title = "Simulated: tertiary-educated foreign-born women")
```



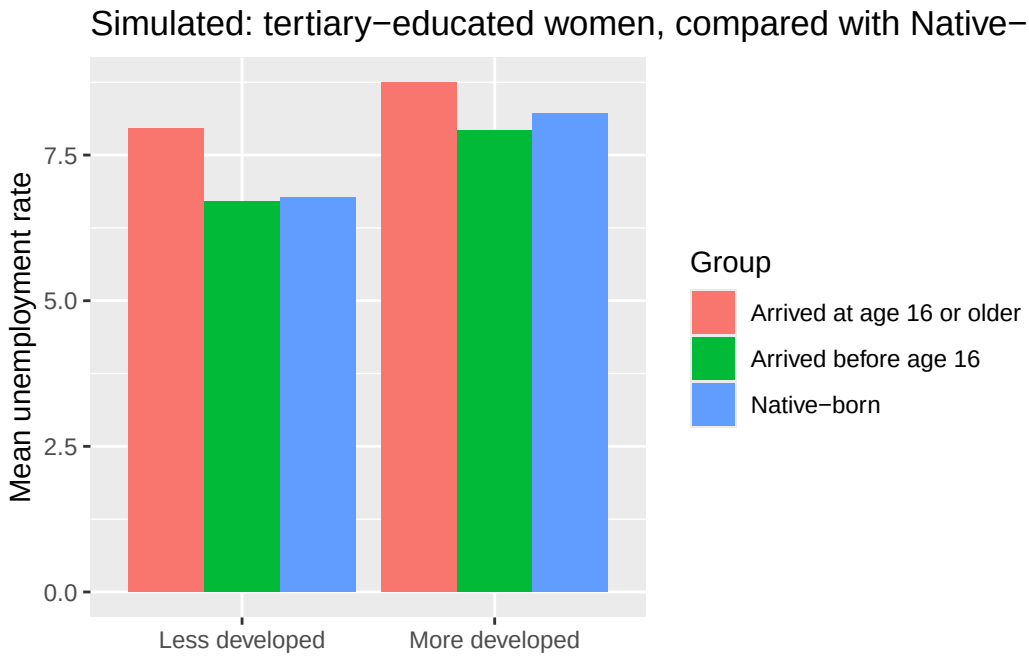

```
mean_by_comparison <-
  women_tertiary_simulated |>
  group_by(development_group, comparison_group) |>
  summarise(mean_rate = mean(unemployment_rate), .groups = "drop")

mean_by_comparison
```

```
# A tibble: 6 x 3
  development_group comparison_group           mean_rate
  <chr>             <chr>                          <dbl>
1 Less developed    Arrived at age 16 or older      7.95
2 Less developed    Arrived before age 16           6.7
3 Less developed    Native-born                     6.77
4 More developed    Arrived at age 16 or older      8.74
5 More developed    Arrived before age 16           7.92
6 More developed    Native-born                     8.21
```

```
mean_by_comparison |>
  ggplot(aes(x = development_group, y = mean_rate, fill = comparison_group)) +
  geom_col(position = "dodge") +
  labs(x = NULL,
       y = "Mean unemployment rate",
```

```
       fill = "Group",
       title = "Simulated: tertiary-educated women, compared with Native-born")
```




```
#mean rate of native-born women
native_born_mean <-
  women_tertiary_simulated |>
  filter(birth_place == "Native-born") |>
  group_by(region, development_group) |>
  summarise(native_born_rate = mean(unemployment_rate), .groups = "drop")

native_born_mean
```

```
# A tibble: 7 x 3
  region           development_group native_born_rate
  <chr>            <chr>                        <dbl>
1 Eastern Europe   Less developed                6.05
2 Latin America    Less developed                6.75
3 Northern America More developed               11.6
4 Northern Europe  More developed                7.99
5 Oceania          More developed                4.9
```

```
6 Southern Europe  Less developed                7.64
7 Western Europe   More developed                8.43
```

```
#mean rate of foreign-born women, by age at arrival
foreign_born_mean <-
  women_tertiary_simulated |>
  filter(birth_place == "Foreign-born") |>
  group_by(region, development_group, arrival_age) |>
  summarise(foreign_born_rate = mean(unemployment_rate), .groups = "drop")

foreign_born_mean
```

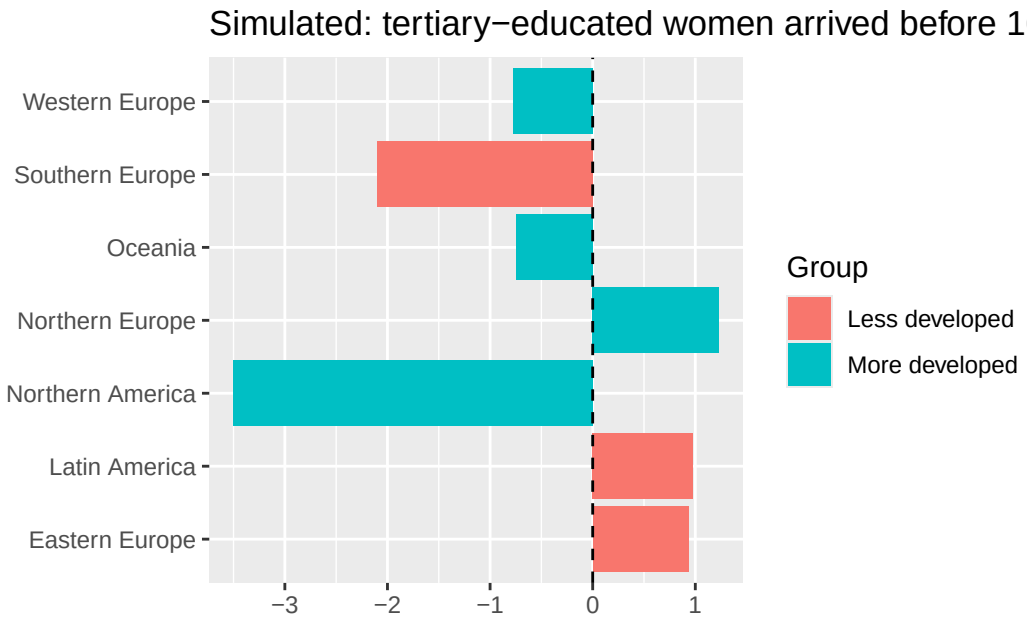
```
# A tibble: 14 x 4
   region           development_group arrival_age              foreign_born_rate
   <chr>            <chr>             <chr>                                <dbl>
 1 Eastern Europe   Less developed    Arrived at age 16 or ol~              9.27
 2 Eastern Europe   Less developed    Arrived before age 16                 6.98
 3 Latin America    Less developed    Arrived at age 16 or ol~              5.58
 4 Latin America    Less developed    Arrived before age 16                 7.72
 5 Northern America More developed    Arrived at age 16 or ol~              7.05
 6 Northern America More developed    Arrived before age 16                 8.05
 7 Northern Europe  More developed    Arrived at age 16 or ol~              8.71
 8 Northern Europe  More developed    Arrived before age 16                 9.21
 9 Oceania          More developed    Arrived at age 16 or ol~             11.2
10 Oceania          More developed    Arrived before age 16                 4.15
11 Southern Europe  Less developed    Arrived at age 16 or ol~              8.28
12 Southern Europe  Less developed    Arrived before age 16                 5.54
13 Western Europe   More developed    Arrived at age 16 or ol~              8.53
14 Western Europe   More developed    Arrived before age 16                 7.66
```

```
#calculate the gap
gap_by_region <-
  women_tertiary_simulated |>
  group_by(region, development_group) |>
  summarise(
    native_born_rate = mean(unemployment_rate[birth_place == "Native-born"]),
    foreign_born_before_16_rate = mean(unemployment_rate[arrival_age == "Arrived before age 1
    foreign_born_16_or_older_rate = mean(unemployment_rate[arrival_age == "Arrived at age 16
    .groups = "drop"
  ) |>
  mutate(gap_before_16 = foreign_born_before_16_rate - native_born_rate,
         gap_16_or_older = foreign_born_16_or_older_rate - native_born_rate)
```

```r
gap_by_region
```

```
# A tibble: 7 x 7
  region           development_group native_born_rate foreign_born_before_16_r~1
  <chr>            <chr>                        <dbl>                       <dbl>
1 Eastern Europe   Less developed                6.05                        6.98
2 Latin America    Less developed                6.75                        7.72
3 Northern America More developed               11.6                         8.05
4 Northern Europe  More developed                7.99                        9.21
5 Oceania          More developed                4.9                         4.15
6 Southern Europe  Less developed                7.64                        5.54
7 Western Europe   More developed                8.43                        7.66
# i abbreviated name: 1: foreign_born_before_16_rate
# i 3 more variables: foreign_born_16_or_older_rate <dbl>, gap_before_16 <dbl>,
#   gap_16_or_older <dbl>
```

```r
gap_by_region |>
  ggplot(aes(x = region, y = gap_before_16, fill = development_group)) +
  geom_col() +
  geom_hline(yintercept = 0, linetype = "dashed") +
  coord_flip() +
  labs(x = NULL,
       y = "Gap: foreign-born (arrived before 16) minus native-born",
       fill = "Group",
       title = "Simulated: tertiary-educated women arrived before 16, GAP")
```




```
gap_by_region |>
  ggplot(aes(x = region, y = gap_16_or_older, fill = development_group)) +
  geom_col() +
  geom_hline(yintercept = 0, linetype = "dashed") +
  coord_flip() +
  labs(x = NULL,
       y = "Gap: foreign-born (arrived 16 or older) minus native-born",
       fill = "Group",
       title = "Simulated: tertiary-educated women arrived over 16, GAP")
```

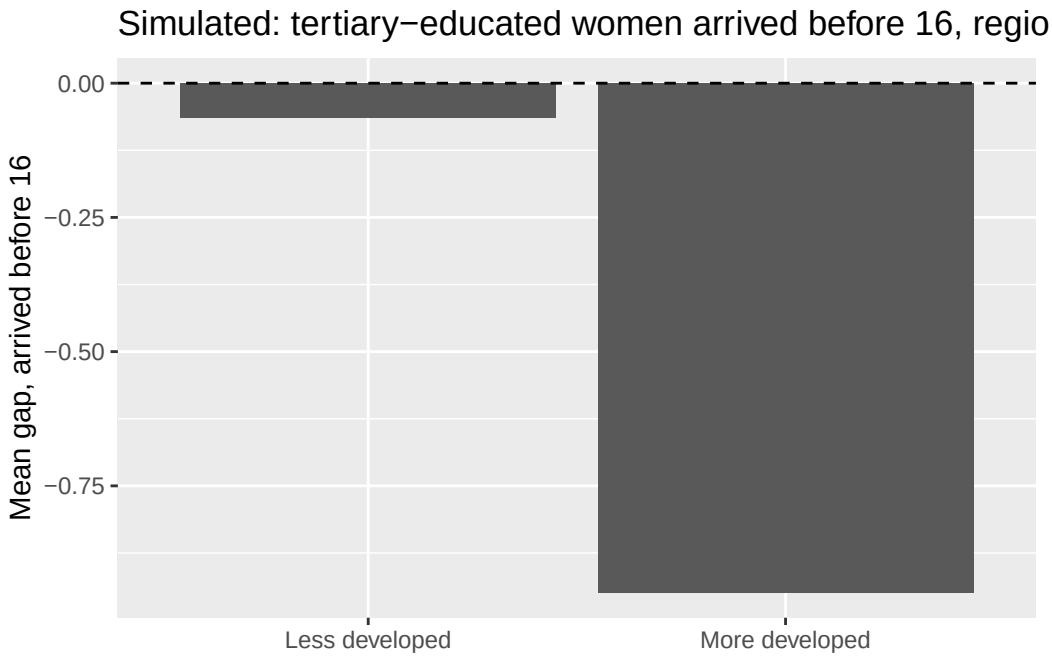



```
gap_by_development <- gap_by_region |>
  group_by(development_group) |>
  summarise(mean_gap_before_16 = mean(gap_before_16),
            mean_gap_16_or_older = mean(gap_16_or_older))

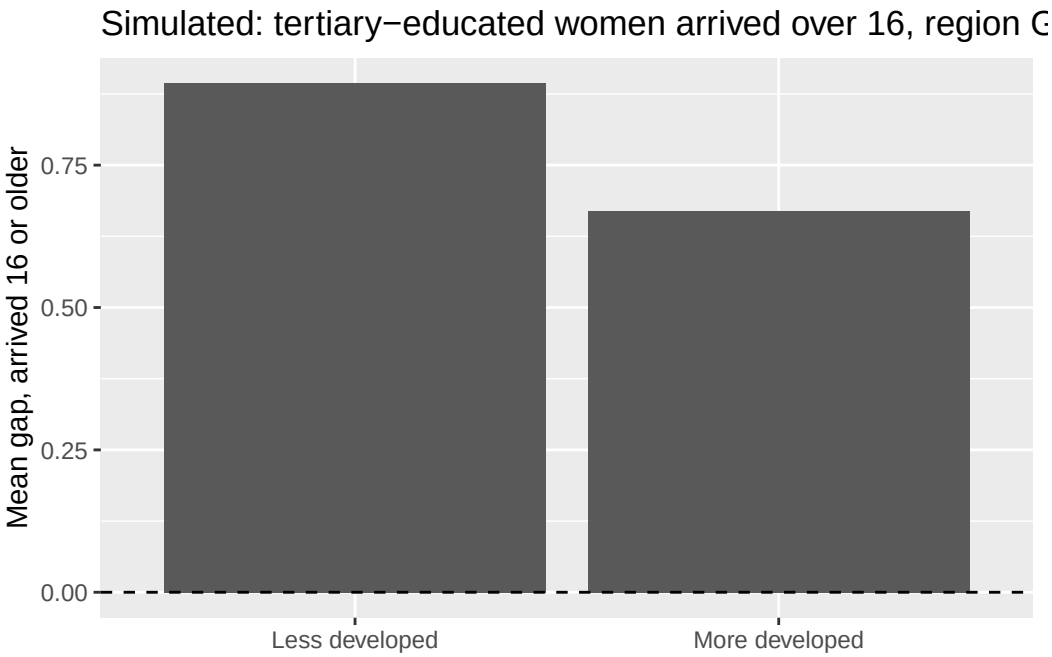
gap_by_development
```

```
# A tibble: 2 x 3
  development_group mean_gap_before_16 mean_gap_16_or_older
  <chr>                          <dbl>                <dbl>
1 Less developed               -0.0639                0.894
2 More developed               -0.948                 0.670
```

```
gap_by_development |>
  ggplot(aes(x = development_group, y = mean_gap_before_16)) +
  geom_col() +
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(x = NULL,
       y = "Mean gap, arrived before 16",
       title = "Simulated: tertiary-educated women arrived before 16, region GAP")
```




```
gap_by_development |>
  ggplot(aes(x = development_group, y = mean_gap_16_or_older)) +
  geom_col() +
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(x = NULL,
       y = "Mean gap, arrived 16 or older",
       title = "Simulated: tertiary-educated women arrived over 16, region GAP")
```


Simulated: tertiary−educated women arrived over 16, region G
Mean gap, arrived 16 or older
0.75
0.50
0.25
0.00
Less developed
More developed